

7(a)	force per unit charge	M1
	(force on) positive charge	A1
7(b)(i)	no electric field inside a conductor	B1
	$R = 4.5 \text{ cm}$	A1
7(b)(ii)	$E = Q / (4\pi\epsilon_0 x^2)$	C1
	clear correct read-off of a pair of values of E and x	C1
	e.g. $Q = 18 \times 10^5 \times 4\pi \times 8.85 \times 10^{-12} \times (4.5 \times 10^{-2})^2$ $= 4.0 \times 10^{-7} \text{ C}$ or $4.1 \times 10^{-7} \text{ C}$	A1
7(c)	At 8.0 cm, $E = 5.75 \times 10^5 \text{ V m}^{-1}$	C1
	$F = Eq$ and $a = F / m$	C1
	$F = (5.75 \times 10^5 \times 2 \times 1.6 \times 10^{-19}) / (4 \times 1.66 \times 10^{-27})$ $= 2.8 \times 10^{13} \text{ m s}^{-2}$	A1